

Course 4: Deep Learning



IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom

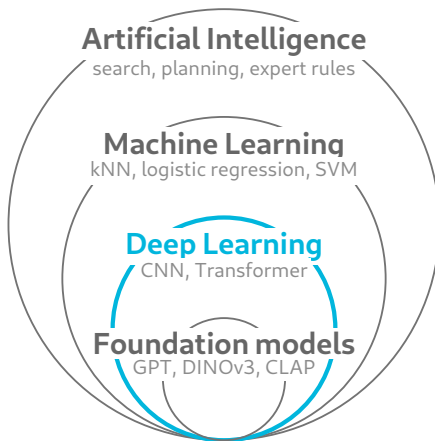
Last session

- 1 Unsupervised learning - discover structure from unlabeled data
- 2 Clustering
- 3 Decomposition
- 4 Manifold learning
- 5 Evaluation
- 6 Feature cleaning

Today's session

- Multi-Layer Perceptron
- Convolutional Neural Networks
- Transformers
- Deep Learning today

Where does Deep Learning sit?



Adapted from *Deep Learning*, 2016 by A. Courville, I. Goodfellow and Y. Bengio

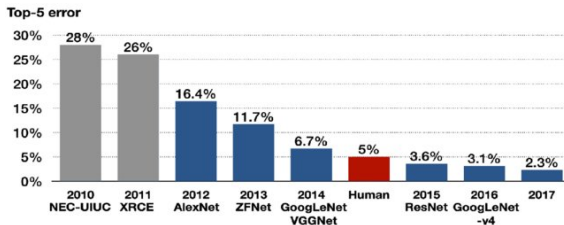
Deep Learning is a **family of models**, not a learning framework: it can be **supervised** (session 2), **unsupervised** (session 3), **self-supervised** (session 5), ...

Deep Learning in a nutshell (1/2)

Tentative definition

- Using *deep* Artificial Neural Networks
- Optimized on *large* datasets.

We generally talk about “Neural Networks” instead of “**Artificial** Neural Networks”, but the latter is the most accurate terminology!

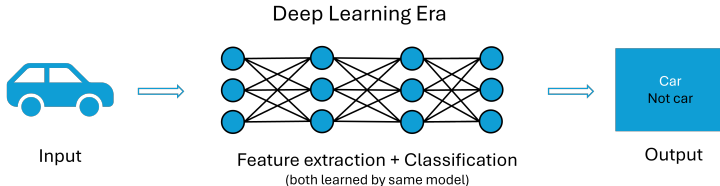
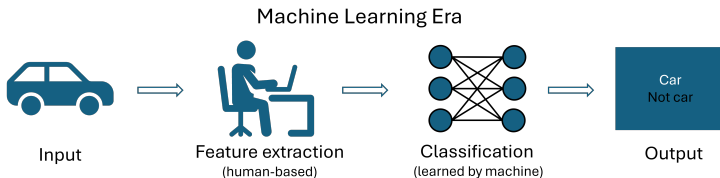


The hype really started in 2012 due to impressive results in image classification.

Source: Kang, D. Y., Duong, H. P., & Park, J. C. (2020). Application of deep learning in dentistry and implantology. *Journal of implantology and applied sciences*, 24(3), 148-181.

Deep Learning in a nutshell (2/2)

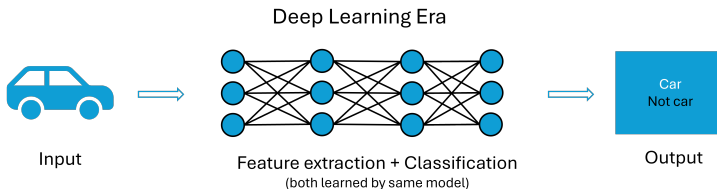
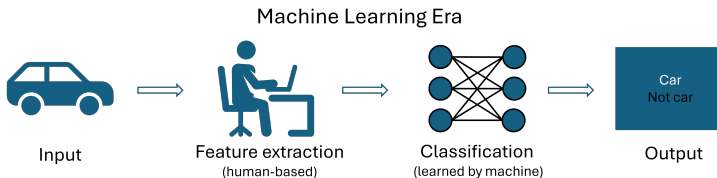
No more hand-crafted features: the **representation is learned**, jointly with the decision.



Inspired from <https://www.softwaretestinghelp.com/data-mining-vs-machine-learning-vs-ai/>

Deep Learning in a nutshell (2/2)

No more hand-crafted features: the **representation is learned**, jointly with the decision.



The price to pay: a lot (a lot) of data.

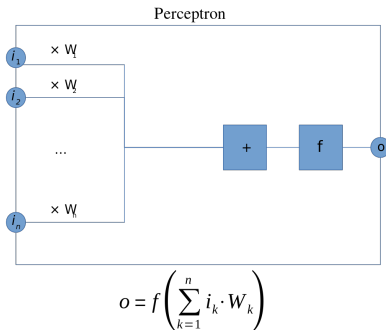
Outline

- 1 Multi-layer Perceptron
- 2 Convolutional Neural Networks (CNN)
- 3 Transformers
- 4 Deep Learning today
- 5 Lab session

Perceptron

1943, first known implementation in 1957

⇒ a weighted sum followed by a nonlinearity **f**; weights **W** are learned.



Source: By Mat the w at English Wikipedia, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=23766733>

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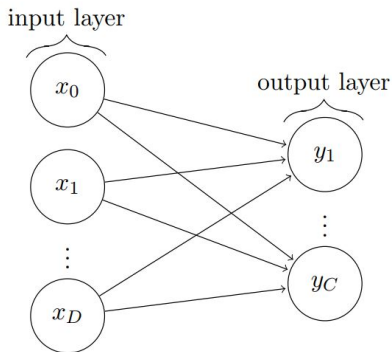


Figure: The arrows represent the weights W , and the circles represent the activation function $\mathbf{f}$.

Optimizing the weights

Loss

- Prediction: $\hat{\mathbf{y}} = f\left(\sum_{d=0}^D x_d W_d\right)$
- Ground truth: $\mathbf{y}$
- Loss (one example): $\mathcal{L}(\hat{\mathbf{y}}, \mathbf{y})$
(ex: $\mathcal{L}(\hat{\mathbf{y}}, \mathbf{y}) = \|\hat{\mathbf{y}} - \mathbf{y}\|_2^2$)
- Loss (i examples): $J(W) = \sum_i \mathcal{L}(\hat{\mathbf{y}}^{(i)}, \mathbf{y}^{(i)})$

Gradient descent

- Compute the gradient: $\frac{\partial J(W)}{\partial W}$ (high dimensional derivative)
- Update weights: $W \leftarrow W - \eta \frac{\partial J(W)}{\partial W}$

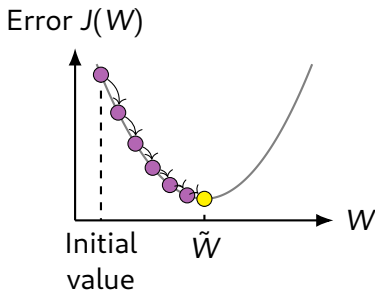
Gradient descent

Intuition behind the gradient descent

Update is given as: $W \leftarrow W - \eta \frac{\partial J(W)}{\partial W}$

- $\frac{\partial J(W)}{\partial W}$ gives the direction
- η gives the size of the step

Well-chosen step: converges to $\tilde{W}$



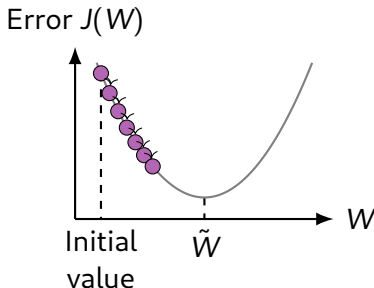
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Too small a step: converges, but very slowly



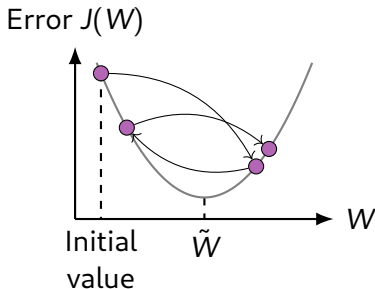
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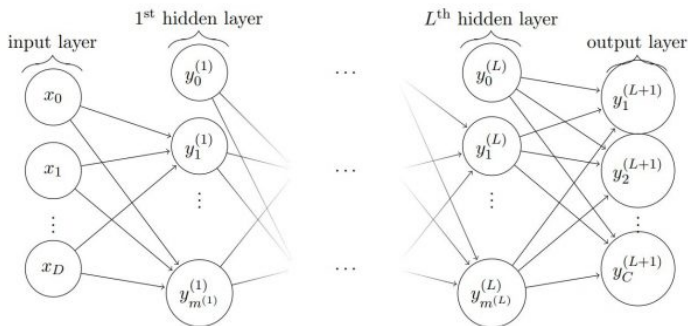
Too large a step: oscillates, may never converge



Multi-Layer Perceptron

Multi-Layer Perceptron (= *fully-connected* network)

- Stacking Perceptrons.
- The **deep** term comes from this stacking
- Prediction: $\hat{\mathbf{y}} = f(W^{(L)} \dots f(W^{(2)} f(W^{(1)} \mathbf{x}))$



Definitions

$$\mathbf{h}^{(l+1)} = f(\mathbf{W}^{(l)}\mathbf{h}^{(l)} + \mathbf{b}^{(l)}) = \phi^{(l)}(\mathbf{h}^{(l)})$$

with $\mathbf{h}^{(0)} = \mathbf{x}$, $\mathbf{h}^{(L)} = \hat{\mathbf{y}}$

- Each building block $\mathbf{h}^{(l+1)}$ is called a **layer**.
- One element i of a layer ($h_i^{(l)}$) is called a **neuron** (both as input and output).
- The nonlinear function f is called the **activation function**.
- $\mathbf{W}^{(l)}$ are called **weights**.
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Note: while each layer can have a different activation function f , it is standard that each layer uses the same.

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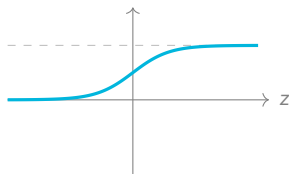
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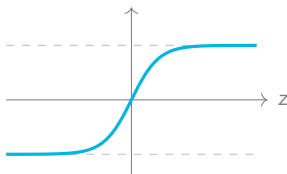
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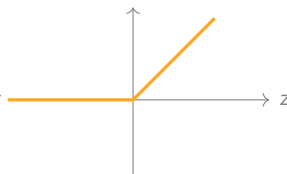
The activation function



sigmoid
 $\sigma(z) = 1/(1 + e^{-z})$



tanh
 $\tanh(z)$



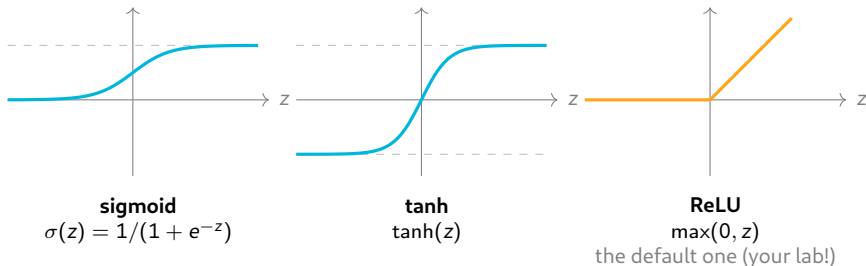
ReLU
 $\max(0, z)$
the default one (your lab!)

Why can we not simply drop f ?

$$W^{(2)} \left(W^{(1)} \mathbf{x} \right) = \underbrace{\left(W^{(2)} W^{(1)} \right)}_{\text{one single matrix}} \mathbf{x}$$

Without f , 100 stacked layers $\approx$ 1 layer: depth would be useless.

The activation function



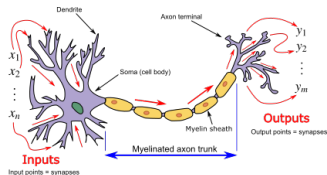
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Why is it called Neural Network?

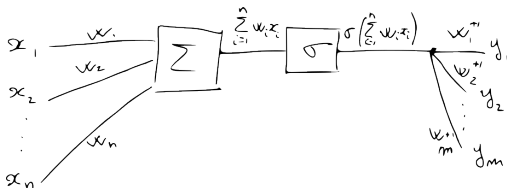
“Neurons” may be seen as **loosely** inspired from the human brain.



Attribution: Egm4313.s12 at English Wikipedia,

[https://en.wikipedia.org/wiki/Neural_network_\(machine_learning\)#/media/](https://en.wikipedia.org/wiki/Neural_network_(machine_learning)#/media/File:Neuron3.png)

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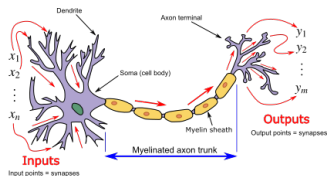
Detail of one neuron in an Artificial Neural Network.

* Late note: I forgot the bias in the graphic.

This is an **analogy**, artificial neural networks are **not** following the human brain (in general).

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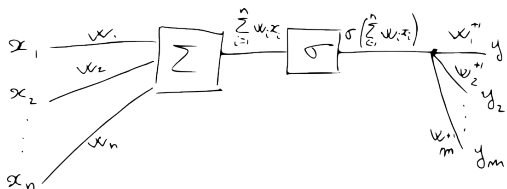
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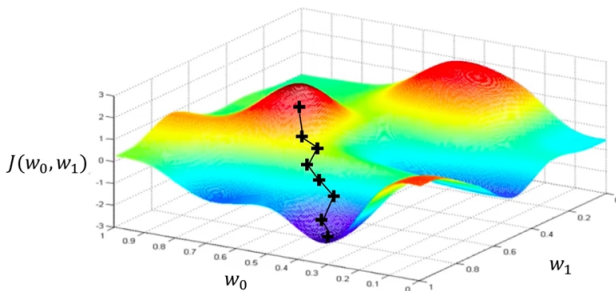
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Gradient descent for deep neural networks

Backpropagation

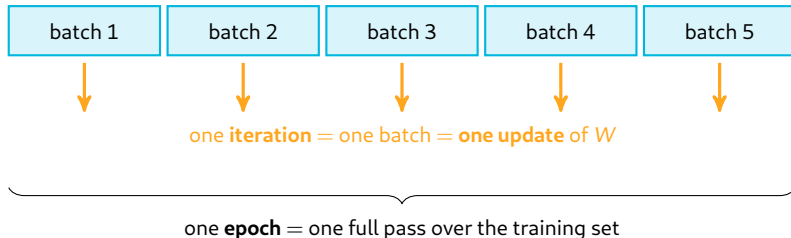
- Gradient descent for all layers (chain rule).
- Simplified equation: $\frac{\partial J(W)}{\partial W} = \frac{\partial J(W)}{\partial W^{(L)}} \frac{\partial W^{(L)}}{\partial W^{(L-1)}} \frac{\partial W^{(L-1)}}{\partial W^{(L-2)}} \dots \frac{\partial W^{(2)}}{\partial W^{(1)}}$
- The error **backpropagates** through the network (reverse path)
- Computationally efficient, but finds a local minimum (at best)



Source: <http://introtodeeplearning.com/>

Batch, iteration, epoch

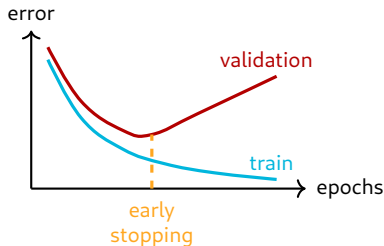
Training set, shuffled and cut into **batches**:



Why not use all the data at once?

- It would not fit in memory,
- One update per epoch would be far too slow.

Overfitting strikes back (Session 2)



Millions of parameters $\Rightarrow$ huge variance

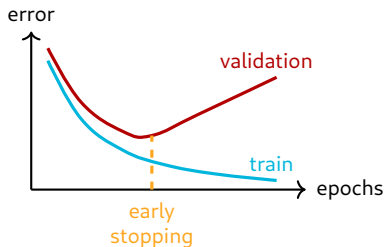
A deep network can **memorize** its training set (even random labels!).

Mitigating overfitting

- **More data** (the best cure),
- **Data augmentation**: rotate/crop/noise,
- **Dropout**: randomly switch off neurons,
- **Early stopping**: keep the best epoch,
- **Weight decay**: penalize large weights.

Always watch the validation curve!

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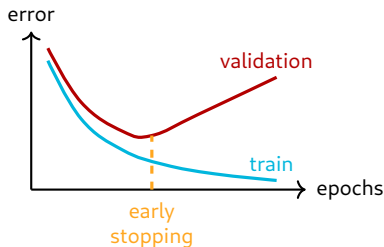
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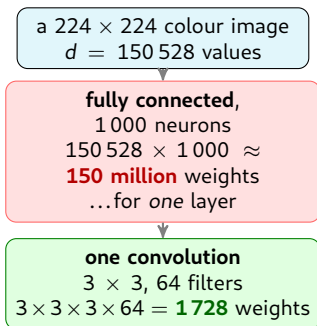
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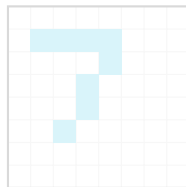
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Limits of the MLP

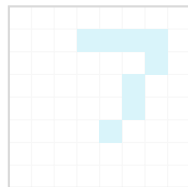
1. Too many parameters



2. No notion of structure



original



shifted by 2 pixels

Same object for you...

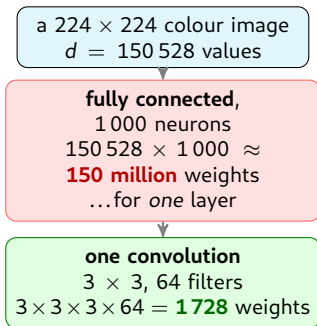
but once **vectorized**, two **very distant** vectors.

⇒ exactly the problem we met on MNIST in Session 3!

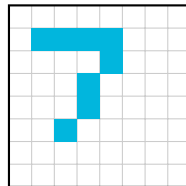
The MLP assumes nothing about the input... which is the problem.

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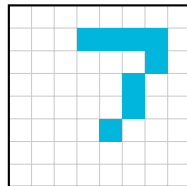
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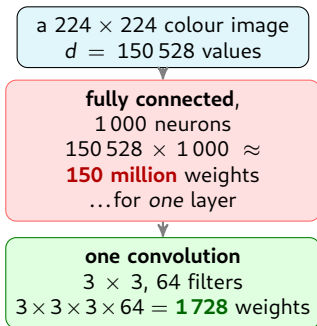
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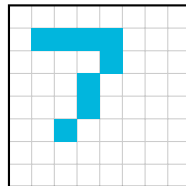
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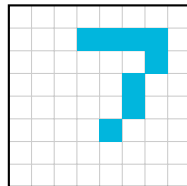
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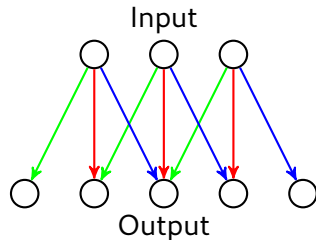
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Convolutional Neural Network (1/5): the principle

Two ideas, motivated for image processing

- **Local:** a kernel looks at a small neighbourhood at a time
⇒ few weights,
- **Shared:** the **same** weights are applied everywhere
⇒ **translation equivariance**.

Convolutional layer



$$\begin{pmatrix} w_1 & w_2 & w_3 & 0 & 0 & 0 \\ 0 & w_1 & w_2 & w_3 & 0 & 0 \\ 0 & 0 & w_1 & w_2 & w_3 & 0 \end{pmatrix}$$

Convolutional Neural Network (2/5): the operation

Example of 2D convolution:

The diagram illustrates the 2D convolution operation. It shows the multiplication of an input matrix I (7x7) by a kernel matrix K (3x3) to produce an output matrix $I * K$ (5x5).

Input matrix I :

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Kernel matrix K :

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

Output matrix $I * K$:

$$\begin{pmatrix} 1 & 4 & 3 & 4 & 1 \\ 1 & 2 & 4 & 3 & 3 \\ 1 & 2 & 3 & 4 & 1 \\ 1 & 3 & 3 & 1 & 1 \\ 3 & 3 & 1 & 1 & 0 \end{pmatrix}$$

The calculation for the value 4 at position (1,4) in the output matrix is shown, corresponding to the 3x3 region of the input matrix highlighted in red:

$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{matrix} \times 1 & \times 0 & \times 1 \\ \times 0 & \times 1 & \times 0 \\ \times 1 & \times 0 & \times 1 \end{matrix} = 4$$

Source: <https://tex.stackexchange.com/questions/437007/drawing-a-convolution-with-tikz>

Convolutional Neural Network (2/5): the operation

Example of 2D convolution:

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} * \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 4 & 3 & 4 & 1 \\ 1 & 2 & 4 & 3 & 3 \\ 1 & 2 & 3 & 4 & 1 \\ 1 & 3 & 3 & 1 & 1 \\ 3 & 3 & 1 & 1 & 0 \end{pmatrix}$$

$I \qquad K \qquad I * K$

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Convolutional Neural Network (3/5): pooling

Example of 2D pooling:

1	2	3	1
1	1	1	1
2	3	1	6
8	1	4	5



maxpool, kernel 2, stride 2

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2	3

maxpool, kernel 2, stride 2

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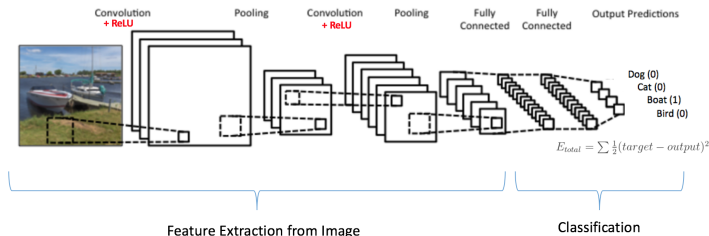
2	3
8	6

maxpool, kernel 2, stride 2

Convolutional Neural Network (4/5): and repeat...

A CNN is mainly *Convolution + Pooling*, repeated

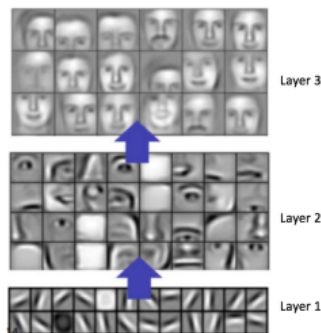
- Progressively: **smaller** maps, **more** channels, Ending with a small MLP to decide.
- ...But many other components may be added! (batch norm, dropout, skip connections, concatenation, ...)
- Not only for images: **1D** convolutions for audio, **3D** for video, ...



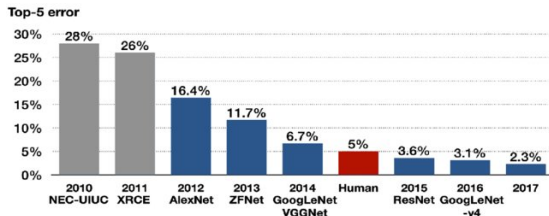
Convolutional Neural Network (5/5): what is learned?

Why convolutions?

- Kernels capture important information in images,
- They become more and more **complex** (and more **abstract**) with the depth.



So, what really happened in 2012?



Not a new idea: a combination of four ingredients

- Convolutional networks (1989!)
- A very large dataset (ImageNet)
- **GPUs** for computation
- Clever tricks (data augmentation, ReLU, dropout)

Source: Kang, D. Y., Duong, H. P., & Park, J. C. (2020). Application of deep learning in dentistry and implantology. *Journal of implantology and applied sciences*, 24(3), 148-181.

Human eval: Russakovsky et al., ImageNet Large Scale Visual Recognition Challenge, <https://arxiv.org/pdf/1409.0575.pdf>

Outline

- 1 Multi-layer Perceptron
- 2 Convolutional Neural Networks (CNN)
- 3 Transformers
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Standard architecture nowadays

- No convolution
- Based on *attention*: what should be important for context?
- Used for text, image, audio, ...

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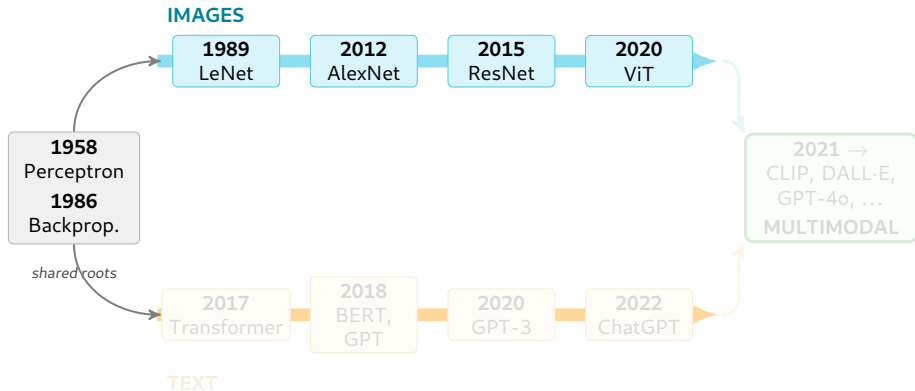
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Will be presented in the next class (foundation models)!

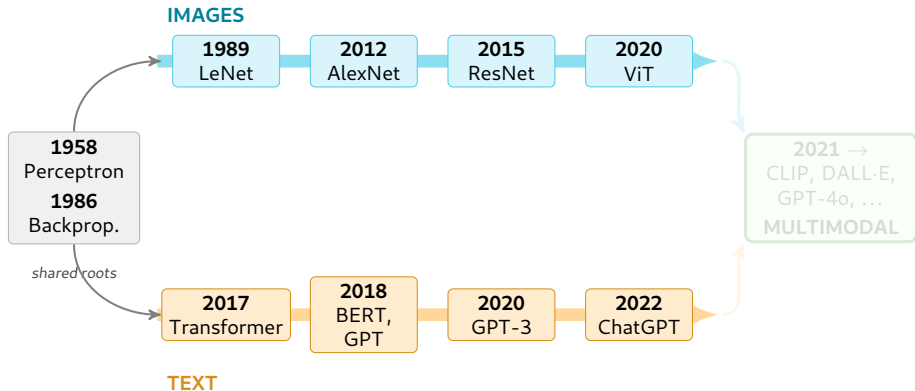
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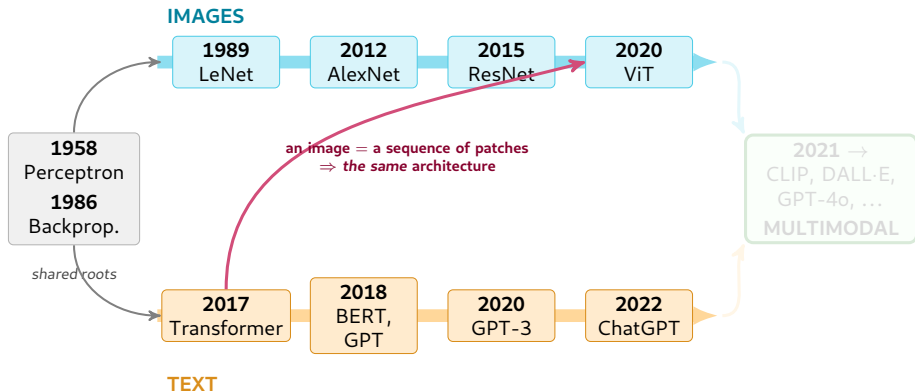
Images and Text drives the AI revolution



Images and Text drives the AI revolution

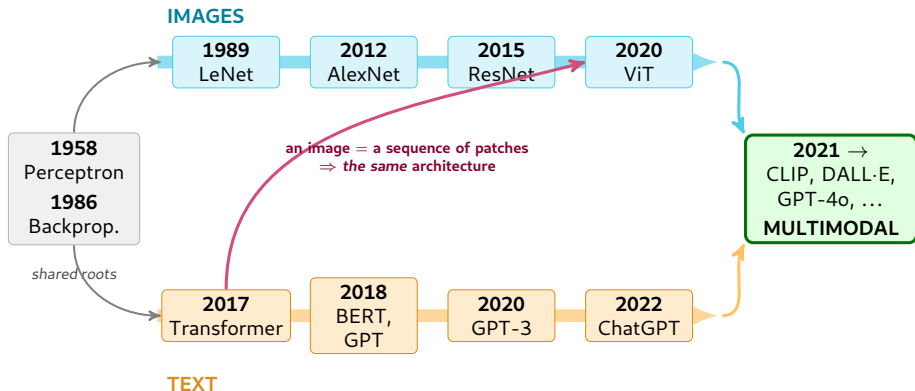


Images and Text drives the AI revolution



An additional note: Recent models all use the Transformer architecture!

Images and Text drives the AI revolution



An additional note: Recent models all use the Transformer architecture!

Models keep growing in size

Number of parameters

- AlexNet (2012): 62 Million
- GPT-3 (2020): 175 Billion

⇒ a factor $\approx 3\,000$ in 8 years.

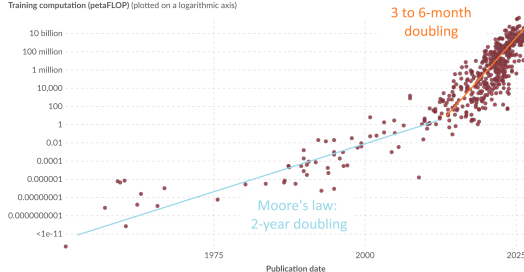
The training cost follows: data, GPU-hours, energy.

Computation used to train notable artificial intelligence systems

Our World In Data

Computation is measured in total petaFLOP, which is 10^{15} floating-point operations¹. Estimated from AI literature, albeit with some uncertainty. Estimates are expected to be accurate within a factor of 2, or a factor of 5 for recent undisclosed models like GPT-4.

Training computation (petaFLOP) (plotted on a logarithmic axis)



Data source: Epoch AI (2026)

OurWorldInData.org/artificial-intelligence | CC BY

¹ Floating-point operation A floating-point operation (FLOP) is a type of computer operation. One FLOP represents a single arithmetic operation involving floating-point numbers, such as addition, subtraction, multiplication, or division.

Computing power used to train AI models, over time (session 1).

Note the **logarithmic** vertical scale.

State of the art: what is solved, what is not

Essentially solved

- Classification on a **large, clean, public** dataset,
- Generating fluent text, plausible images, speech,
- Building a good **generic representation** of images, text, audio.

Still hard

- **Little data**, or a **specialized** domain (medical imaging, industry),
- **Transferring** to a dataset that differs from the training one,
- **Reliability**: hallucinations, robustness, calibration,
- **Cost**: data, energy, hardware.

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What to remember: an architecture is a *prior*

	MLP	CNN	Transformer
<i>sees the data as...</i>	 a flat vector	 a regular grid	 a set of tokens
<i>key idea</i>	everything is connected to everything	local window, weights shared everywhere	attention : the neighbourhood is learned
<i>consequence</i>	huge number of parameters, no invariance	few parameters, translation invariance	very general, but very data-hungry
<i>typical data</i>	tabular data, embeddings	images, audio, video	text, and... everything else today

Choosing an architecture = choosing what you already know (session 2!).

Conclusion

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- **Powerful without feature extraction:** the representation is **learned**,
- They require **a lot of data** to be trained,
- **The architecture plays an important role:** it encodes what you know about your data.

Common criticisms

- Hard to **interpret**,
- **Reproduce the biases** present in the data,
- May require massive amounts of **energy**.

Going further

- Details and maths behind AI (3Blue1Brown): <https://youtu.be/aircAruvnKk>
- Intelligence and AI (M.Phi, french): <https://youtu.be/AwUpxj0DogQ>
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Lab Session 4: Deep Learning with PyTorch

Part 1 — The training loop, with PyTorch

- **Tensors:** like `numpy` arrays, but with gradients and GPU support,
- Define a model (`nn.Module`, `nn.Linear`, `nn.Conv2d`, `ReLU`, pooling),
- The **training loop**, and it is always the same five lines: `zero_grad` → `forward` → `loss` → `backward` → `step`,
- **Train a small CNN** on CIFAR-10, and evaluate it (accuracy, per class),
- **Look inside:** display the feature maps of the first convolutional layer.

Part 2 — Your own project data

Take the same training loop, and train a first model on your project.
⇒ what is one sample, and what shape does it have? which loss? which final layer? Keep a **validation set**, plot train and validation loss, and compare with the `sklearn` baselines of sessions 2 and 3.

Link to the lab: <https://mee-labs.gitlab-pages.imt-atlantique.fr/intro2ai/lab4/>

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